

Cellular Biophysics (BIO516) – Final Sem

Marks: 40, Time: 2 hr

Date: 29/04/2026

Instructions to candidates

1. This paper consists of four sections: Section A (Multiple Choice: Q1-Q10), Section B (Fill in the Blanks: Q11-Q20), Section C (Subjective Questions: Q21-Q26), and Section D (Numerical Problems: Q27-Q31).
2. **Answer all the questions**
3. Use of calculator is permitted.
4. Standard Constants:
 - Gas Constant (R) = 8.315 J/(mol · K)
 - Faraday's Constant (F) = 96,485 C/mol
 - Temperature (T) = 298 K (unless specified)

Section A

[Marks: 10 X 1]

Q1 Assertion: The sodium channel I_1 state is considered a "non-conducting" but "active-gate open" state. **Reason:** The inactivation gate blocks the pore even when all three m-gates (activation gates) are in the open position.

- A) **Both Assertion and Reason are true, and Reason is the correct explanation.**
B) Both are true, but Reason is not the explanation.
C) Assertion is false.
D) Reason is false.

Q2 According to Rudnick's rules, the serotonin transporter can flip from the extracellular side to the intracellular side ONLY when:

- A) Only sodium is bound.
B) Sodium, chloride, and serotonin are all bound.
C) The binding site is empty.
D) Only potassium is bound.

Q3 According to the simplest model of alternating access carrier transport, which of the following statements about the carrier transition rate constant (D') is true?

- A) It strictly depends on whether the carrier is facing Side I or Side II, creating an inherent asymmetry.
B) It is assumed to be the same for transitions in either direction, and identical for both the loaded and unloaded carrier.
C) It is always much higher for the loaded carrier, preventing the carrier from transitioning when empty.
D) It relies directly on ATP hydrolysis to drive the conformational transition from Side I to Side II.

Q4 In a secondary active cotransport (symport) system transporting substrate A alongside cosolute B, what does the phenomenon of "cis stimulation" describe?

- A) Raising the concentration of A on the trans side increases the rate at which B crosses.
B) The independent movement of B across the membrane triggering a conformational change in an adjacent channel.
C) Raising the concentration of B in Phase I drives the initial binding reaction, increasing the active flux of A from Phase I to Phase II.

D) Lowering the electrochemical gradient of B forces the pump to act as an antiporter.

Q5 Which enzyme is responsible for the termination of the β -adrenergic response by degrading cAMP?

A) Adenylyl cyclase

B) Protein Kinase A

C) Cyclic nucleotide phosphodiesterase

D) Phosphatase

Q6. In voltage-gated sodium channels, what structural element is directly responsible for sensing changes in the transmembrane potential?

A) The peptide backbone carbonyl oxygens located in the selectivity filter.

B) A high density of positively charged Arginine (Arg) residues on Helix 4 (S4).

C) The tethered segment connecting domains III and IV.

D) The lipid-facing hydrophobic residues on Helix 6 (S6).

Q7 What mechanism dictates whether the spatial and temporal summation of Excitatory Postsynaptic Potentials (EPSPs) will successfully trigger an action potential in a postsynaptic neuron?

A) The combined electrotonic spread of depolarization reaching the axon hillock and exceeding its specific excitation threshold.

B) The immediate opening of local voltage-gated chloride channels in the dendrite.

C) The complete phosphorylation of all available metabotropic receptors in the soma.

The release of a single quantum of neurotransmitter precisely aligned with an active zone.

Q8 In the ball-and-chain model of ion channel inactivation, the "ball" that blocks the pore is a:

A) Free-floating ion

B) Tethered segment of the protein

C) Lipid molecule

D) Sugar moiety

Q9 You are a student in my lab, in a "Patch-Clamp" experiment, if you wish to study the effect of intracellular second messengers (like cAMP or Ca^{2+}) on an ion channel while having easy access to change the medium "bathing" the cytoplasmic surface, which configuration should you use?

A) Cell-attached mode

B) Whole-cell mode

C) Inside-out patch

D) Outside-out patch

Q10 The fluorescence mechanism of the Fluo-4 calcium indicator relies on Photoinduced Electron Transfer (PeT). What exactly happens when Ca^{2+} binds to the BAPTA moiety of Fluo-4?

A) BAPTA's reduction potential increases, making it a better electron donor to the fluorophore.

B) BAPTA undergoes a conformational shift that lowers its reduction potential, blocking the PeT process and allowing the xanthene moiety to emit fluorescence.

C) Fluo-4 undergoes non-radiative energy transfer to a nearby GFP molecule via FRET.

D) The Ca^{2+} ion directly absorbs the 488 nm laser light and undergoes an energy shift.

Section B

[Marks: 10 X 1]

Q11 Receptors that are closely associated with channels and participate in slow synaptic transmission are called _____. (Ans: **metabotropic**)

Q12 The stoichiometry of transport, or the number of co-solute molecules that must accompany each molecule of substrate, is known as the _____. (Ans: **coupling coefficient**)

Q13 In FRET-based imaging of cAMP, the FRET signal is _____ when cAMP rises because the R₂C₂ tetramer of PKA dissociates, increasing the distance between the donor and acceptor. potential for that ion. (Ans: **lost/ceases (Any variation of this)**)

Q14 In secondary active transport, when raising the concentration of substrate A on one side of the membrane strictly increases the flux of the cosolute B, the kinetic phenomenon is termed _____ stimulation. (Ans: **reciprocal**)

Q15 Since exocytosis at a synapse is a vesicular transport process where neurotransmitters are released in discrete, uniform packets, the synaptic release is formally described as being _____. (Ans: **quantized**)

Q16 The selectivity filter of a potassium channel perfectly accommodates K⁺ ions by using the precise geometric arrangement of the peptide backbone's _____ oxygens to replace the ion's natural waters of hydration. (Ans: **carbonyl**)

Q17 The Na⁺ – Ca²⁺ antiporter in the plasma membrane utilizes the transmembrane free energy difference of three sodium ions moving down their gradient to push a single calcium ion out of the cell. This overall mechanism is classified as _____ active transport. (Ans: **secondary**)

Q18 In a heavily arborized neuron, action potentials almost always originate physically at the _____ because this specific structural region possesses the lowest threshold for excitation. (Ans: **axon hillock**)

Q19 In the high-Mg²⁺ environment of biological cell fluids, ATP predominantly exists as a stable complex form known as _____, which is the actual species that participates in enzymatic phosphate transfer reactions. (Ans: **Mg.ATP²⁻ (or MgATP²⁻)**)

Q20 In the patch-clamp technique, applying light suction to the fine glass micropipette against a cell membrane creates a high-resistance electrical seal, commonly known as a _____. (Ans: **gigaseal**)

Section C

Q21 The operating curve of an active transport system that pumps A from Phase I to Phase II is approximated by a simple linear expression

$$J_A^a = a + b\Delta c_A$$

(a) In terms of a and b, what is the flux at level flow?

(b) In terms of a and b, what is the concentration difference across the membrane at static head?

[Mark: 1]

Ans: (a) The flux at level flow is when the chemical gradient and hence concentration gradient of the solute across the barrier is 0. This corresponds to flux as “a”.

(b) at static head, the flux of the solute across the barrier is 0. So, concentration gradient is “-a/b”.

Q22 An ion substitution experiment is carried out on a nerve axon, in which some of the extracellular sodium is replaced by an equimolar amount of potassium.

(a) What is the effect of the substitution on the resting potential of the axon? Why? (**Given:** $U_{Na} < U_K$; U = mobility of ion). [Mark: 1.5]

Hint: Use the following table to get an idea about concentration gradient of ions. You can also consider the mobility of leakage ions (e.g., Cl^-) as 0.

Species	Axoplasmic concentration (c_i^I), mM	Extracellular concentration (c_i^{II}), mM	Nernst potential, ^a mV
Na	14	142	62
K	140	4	-95
Cl	4	103	-87

^aBased on concentrations. The cell potential is ca. -90 mV. $T = 37^\circ C$.

Ans:

The resting membrane potential is represented by

$$\Delta\psi_r = \frac{RT}{\delta} \ln \frac{U_{Na}c_{Na}^{II} + U_Kc_K^{II} + U_{Cl}c_{Cl}^I}{U_{Na}c_{Na}^I + U_Kc_K^I + U_{Cl}c_{Cl}^{II}}$$

Normally (before substitution):

K^+ concentration is high inside the axon and low outside.

Na^+ concentration is high outside and low inside.

Because K^+ has greater mobility, it tends to diffuse outward more easily than Na^+ diffuses inward. This outward movement of positive charge leaves the inside of the axon negative, creating the resting potential.

When extracellular Na^+ is replaced by K^+ :

Extracellular K^+ concentration increases. This reduces the concentration gradient for K^+ across the membrane. Therefore, less K^+ diffuses out of the cell. Since K^+ dominates the resting potential. Reduced outward movement of K^+ means less positive charge leaves the cell. The inside becomes less negative.

Reduction in Na^+ has smaller effect Because Na^+ mobility is lower than K^+

Final Answer

The resting potential decreases in magnitude and becomes less negative (depolarization) when some extracellular Na^+ is replaced by K^+ . This happens because increasing external K^+ reduces the outward diffusion of K^+ , and since potassium has higher mobility than sodium, K^+ has the dominant influence on the resting membrane potential.

For example, using U_{Na}/U_K as 0.03442

Resting membrane potential according to the composition given in the table is -90 mV.

When 12 mM extracellular Na^+ is substituted by K^+ ions, i.e., Na^+ , out = 130 mM, K^+ , out = 16 mM, then resting membrane potential is -51.5 mV

Q23 A useful parameter of voltage-dependent gating is $\Delta\Psi_{0.5}$, the membrane potential at which half of the channels are open. Use the following equation to relate $\Delta\Psi_{0.5}$ to the standard free energy change of channel opening (ΔG^0).

$$\frac{N_o}{N_T}(\Delta\psi) = \left[1 + \exp\left(\frac{\Delta G^0}{RT}\right) \exp\left(-\frac{nF\Delta\psi}{RT}\right) \right]^{-1}$$

[Marks: 1.5]

Ans: $\Delta\psi_{0.5} = \frac{\Delta G^0}{nF}$

Q24 Outline the cascade from G_q -coupled receptor activation to calcium release. [Mark: 2]

Answer: Hormone binding activates $G_{q\alpha}$, which activates membrane-bound Phospholipase C (PLC). PLC cleaves PIP_2 into DAG and IP_3 . IP_3 diffuses to the endoplasmic reticulum, binding to IP_3 -gated channels and releasing sequestered Ca^{2+} into the cytosol.

Q25 How would you use FRET between green fluorescent protein (excitation: 380 nm, emission: 545 nm) and blue fluorescent protein (excitation: 380 nm, emission: 460 nm) to demonstrate the interaction between β -arrestin and the β -adrenergic receptor in living cells? Which wavelengths would you use to illuminate the cells, and which would you monitor? What would you expect to observe if the interaction occurred? If it did not occur? [Marks: 2]

Ans: 1. Construct design

For example:

β -adrenergic receptor-BFP

β -arrestin-GFP

(These can also be reversed.)

Here, BFP acts as the donor fluorophore (as emission at a shorter wavelength, i.e., higher energy) and GFP acts as the acceptor fluorophore (as emission at a longer wavelength, i.e., lower energy).

2. Principle of the experiment

If β -arrestin binds to the activated β -adrenergic receptor, the two fluorophores are brought within ~1–10 nm of each other, allowing FRET to occur.

When the donor (BFP) is excited, its energy is transferred non-radiatively to GFP, causing GFP fluorescence.

3. Expected results

If interaction occurs (β -arrestin binds receptor)

FRET takes place:

Upon illumination at 380 nm, the FRET results a decrease in BFP emission at 460 nm, and increase in GFP emission at 545 nm. So, after stimulating the receptor (for example with adrenaline/isoproterenol), you would observe a shift in fluorescence from blue to green.

If interaction does not occur

No FRET:

BFP and GFP both are excited upon illumination at 380 nm. As there is no FRET, both exhibit their characteristic emission at 460 nm and 545 nm, respectively.

Q26 Discuss the major difficulties in solving the Schrödinger equation for multi-atom, multi-electron systems and why the role of electron density $n(r)$ in Density Functional Theory is important to reduce the complexity? [Marks: 2]

Ans: Solving the Schrödinger equation for multi-atom, multi-electron systems is difficult because the many-electron wavefunction depends on the coordinates of all electrons simultaneously, making it a high-dimensional problem whose complexity increases rapidly with the number of particles. As a result, exact solutions are practically impossible for large atoms, molecules, or solids. Density Functional Theory (DFT) reduces this complexity by using the electron density $n(r)$ instead of the full

wavefunction. Since $n(r)$ depends only on three spatial coordinates, regardless of the number of electrons, it greatly simplifies the problem while still containing enough information to determine ground-state properties, making calculations of complex systems computationally feasible.

Section D

Q27 The difference in pH between the internal and external surfaces of the inner mitochondrial membrane is 1.4 pH units (external side acidic). If the membrane potential is 0.06 V (inside negative), what is the free energy change on transporting 1 mol of protons across the membrane from outside to inside at 25 °C? [Marks: 2]

Ans:

Free Energy Change for Proton Transport

Given:

$$\Delta \text{pH} = 1.4 \quad (\text{outside more acidic})$$

$$\Delta \psi = \psi_{\text{in}} - \psi_{\text{out}} = -0.06 \text{ V} \quad (\text{inside negative})$$

$$T = 298 \text{ K}$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$F = 96485 \text{ C mol}^{-1}$$

Equation:

$$\Delta G = RT \ln([H^+]_{\text{in}}/[H^+]_{\text{out}}) + zF\Delta\psi$$

Chemical term:

$$RT \ln(10^{-1.4}) = -7.98 \text{ kJ mol}^{-1}$$

Electrical term:

$$(1)(96485)(-0.06) = -5.79 \text{ kJ mol}^{-1}$$

Total:

$$\Delta G = -7.98 + (-5.79)$$

$$\Delta G = -13.77 \text{ kJ mol}^{-1}$$

Final Answer:

Free energy change for transport of 1 mol H^+ from outside to inside is:

$$\Delta G = -13.8 \text{ kJ/mol}$$

$$** \ln([H^+]_{\text{in}}/[H^+]_{\text{out}}) = 2.303 \log([H^+]_{\text{in}}/[H^+]_{\text{out}}) = 2.303(\log[H^+]_{\text{in}} - \log[H^+]_{\text{out}}) = -2.303(-\log[H^+]_{\text{in}} + \log[H^+]_{\text{out}})$$

$$= -2.303\Delta \text{pH (inside-outside)} = -2.303 \times 1.4 = -3.2242$$

$$\text{So, chemical term} = RT(-3.2242) = -7.98 \text{ kJ/mol}$$

Q28 Suppose you determined experimentally that a cellular transport system for glucose, driven by symport of Na, could accumulate glucose to concentrations 25 times greater than in the external medium, while the external [Na] was only 10 times greater than the intracellular [Na]. Would this violate the laws of thermodynamics? If not, how could you explain this observation?

Given: membrane potential of 0.070 V (inside negative).

[Marks: 2]

Ans:

Na⁺-Glucose Symport Free Energy Calculation

No, this does not violate thermodynamics. A Na⁺-glucose symporter uses the electrochemical gradient of Na⁺ to drive glucose uptake.

For Na⁺ transport (outside to inside):

$$\Delta G_{\text{Na}} = RT \ln([Na^+]_{\text{in}}/[Na^+]_{\text{out}}) + zF\Delta\psi$$

Given:

$$[Na^+]_{\text{out}}/[Na^+]_{\text{in}} = 10$$

$$T = 298 \text{ K}$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$z = +1$$

$$F = 96485 \text{ C mol}^{-1}$$

$$\Delta\psi = -0.070 \text{ V}$$

$$\begin{aligned}\Delta G_{\text{Na}} &= 8.314 \times 298 \times \ln(1/10) + (96485)(-0.070) \\ &= -5704 - 6754 \\ &= -12458 \text{ J mol}^{-1} \\ \Delta G_{\text{Na}} &= -12.46 \text{ kJ mol}^{-1}\end{aligned}$$

For glucose transport:

$$\Delta G_{\text{glucose}} = RT \ln([G]_{\text{in}}/[G]_{\text{out}})$$

$$[Glucose]_{\text{in}}/[Glucose]_{\text{out}} = 25$$

$$\begin{aligned}\Delta G_{\text{glucose}} &= 8.314 \times 298 \times \ln(25) \\ &= 7974 \text{ J mol}^{-1} \\ \Delta G_{\text{glucose}} &= +7.97 \text{ kJ mol}^{-1}\end{aligned}$$

Overall symport (1 Na⁺ : 1 glucose):

$$\begin{aligned}\Delta G_{\text{total}} &= \Delta G_{\text{Na}} + \Delta G_{\text{glucose}} \\ &= -12.46 + 7.97 \\ &= -4.49 \text{ kJ mol}^{-1}\end{aligned}$$

Conclusion:

Since ΔG_{total} is negative, the coupled transport is spontaneous.

Thus, glucose accumulation to 25 times outside concentration is thermodynamically allowed due to both the Na⁺ concentration gradient and the inside-negative membrane potential.

Q29 A target cell possesses $N_T = 2.0 \times 10^4$ cell-surface receptors for a specific hormone. The equilibrium dissociation constant for the hormone-receptor complex is $K = 25$ nM. Assuming rapid equilibrium of hormone binding to the receptors, calculate the number of hormone-bound receptors in the cell at steady state when exposed to an ambient hormone concentration of 75 nM.

[Marks: 2]

Ans:

$$N_b = N_T \frac{c_A}{K + c_A} = (2 \times 10^4) \frac{75}{25 + 75} = (2 \times 10^4) \times 0.75 = 15,000$$

Q30 A calcium "sparklet" has a signal mass of $0.34 \mu\text{m}^3$. A $1.0 \mu\text{m}^3$ signal equals exactly 2×10^{-20} moles of Ca^{2+} . If the channel was open for 5.0 ms, calculate the average single-channel Ca^{2+} current in picoamperes (pA). ($F = 96485$ C/mol). [Marks: 2]

Answer: Total moles = $0.34 \times (2 \times 10^{-20}) = 0.68 \times 10^{-20}$ moles. Total Charge (Q) = $(0.68 \times 10^{-20}) \times 2 \times 96485 = 1.312 \times 10^{-15}$ C. Current (I) = 1.312×10^{-15} C / 0.005 s = **0.262 pA**.

Q31 Förster Resonance Energy Transfer is used to estimate molecular distance changes during protein conformational switching. Cyan Fluorescent Protein (CFP) acts as the donor and Yellow Fluorescent Protein (YFP) acts as the acceptor.

A CFP-YFP biosensor exists in two states:

Open state: FRET efficiency = 0.28

Closed state: FRET efficiency = 0.72

FRET efficiency is denoted by:

$$E = \frac{1}{1 + \left(\frac{r}{R_0}\right)^6}$$

For the CFP-YFP pair, the Förster distance (R_0) is 5.2 nm.

$r \rightarrow$ donor-acceptor distance

Calculate:

- (A) The donor-acceptor distance in the open state
- (B) The donor-acceptor distance in the closed state
- (C) The change in distance after closing

[Marks: 2]

Ans:

Formula:

$$E = 1 / (1 + (r/R_0)^6)$$

$$\text{Rearranged: } r = R_0 ((1-E)/E)^{1/6}$$

1. Open State ($E = 0.28$)

$$r = 5.2 \times ((0.72/0.28))^{1/6}$$

$$r = 5.2 \times (2.571)^{1/6}$$

$$r \approx 5.2 \times 1.171 = 6.09 \text{ nm}$$

2. Closed State ($E = 0.72$)

$$r = 5.2 \times ((0.28/0.72))^{(1/6)}$$

$$r = 5.2 \times (0.389)^{(1/6)}$$

$$r \approx 5.2 \times 0.854 = 4.44 \text{ nm}$$

3. Distance Change

$$\Delta r = 6.09 - 4.44 = 1.65 \text{ nm}$$

Final Result

- Open-state distance = 6.09 nm
- Closed-state distance = 4.44 nm
- Distance decrease on closing = 1.65 nm